

Anders von Mirbach

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Computer Science graduate with experience in Python, SQL, PostgreSQL, backend development, and data-focused software systems. Strong foundation in database design, performance benchmarking, testing, and building data-driven applications. Interested in machine learning, data engineering, analytical software, and scalable systems that turn complex data into useful insights.

EDUCATION

University of Lethbridge, Calgary | *B.Sc. Computer Science, April 2026*

Coursework: Operating Systems, Programming Languages, Database Management Systems, Data Analytics. Focus on software development, databases, and AI with applied project work.

S.A.I.T., Calgary | *Diploma, Full-Stack Development, August 2024*

Fast-paced program covering full-stack development, Java and Python programming, and database design. Practical experience through collaborative projects and a capstone.

PROJECTS

Learned Index Structures | *Python · NumPy · B-Tree · RMI*

- Implemented and benchmarked machine-learning-based index structures inspired by Kraska et al. (SIGMOD 2018), including a Recursive Model Index compared against traditional B-Trees.
- Generated large datasets and measured lookup latency, memory usage, prediction error, data distribution effects, and performance trade-offs.

Cannabiz POS | *React · Node.js · Express · PostgreSQL*

- Developed a full-stack point-of-sale system supporting inventory management, order processing, transaction tracking, customer records, and reporting workflows.
- Designed relational database models and backend workflows suited for day-to-day retail operations and structured business data.

SKILLS

Languages: Python, SQL, JavaScript, Java, C/C++, VBA

Machine Learning & Data Analysis: NumPy, pandas, Matplotlib, data generation, benchmarking, performance analysis, model-based indexing

Databases & Backend: PostgreSQL, MySQL, database design, relational data modeling, REST APIs, Node.js, Express

Web Development: React, Next.js, HTML/CSS, full-stack application development

Systems & Tools: Git, GitLab, VS Code, Linux terminal, Excel/VBA, debugging, testing, technical documentation

Computer Science Concepts: Data structures, algorithms, operating systems, database systems, systems programming

EXPERIENCE

Four20, Calgary | *Assistant Manager, December 2022 – Present*

- Manage daily store operations while balancing leadership responsibilities alongside full-time academic commitments.
- Simplified product tag generation by developing Excel/VBA automation that integrates with Word mail-merge templates, enabling staff to produce standardized tags with a single click.

Kymand Incorporated, Calgary | *Team Lead, January 2018 – March 2020*

- Directed a small, high-performing sales team in a fast-paced environment and coached team members to consistently exceed sales targets.

TECHNICAL STRENGTHS

- Strong interest in machine learning, data engineering, analytical systems, and applied AI.
- Comfortable testing and comparing technical approaches using structured datasets and measurable results.
- Experienced documenting project requirements, implementation details, testing methods, and findings.
- Able to connect technical work with practical business and operational needs through prior leadership experience.